

Mustafa Talha Avcu

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WORK EXPERIENCE

- **Amazon** Seattle, WA - Toronto, ON
Software Development Engineer II *August 2022 - June 2026*
 - Helped Macy's onboarding to Amazon's **Retail Ad Service (RAS)**, resolving scaling related catalog ingestion and service throttling issues by improving batching logic, retry strategies, and observability; enabled stable integration for the largest retailer to date.
 - Extended existing RAS attribution and catalog services to integrate with Amazon Demand-Side Platform (ADSP) — enabling advertisers to run display and video campaigns across retailer sites— by migrating AWS Clean Room collaboration to Spark engine, and implementing unified data pipelines and attribution logic across Sponsored Products (SP) and ADSP (expanding from SP-only support).
 - Delivered cross-team support for science teams to enable development and deployment of LLMs in **MessageUs**, Amazon's customer service chatbot; played a key role in launching Amazon's first LLM-powered support experience for Amazon Music-related customer issues.
 - Built a concession feature allowing customers to return out-of-policy Prime Video orders via MessageUs.
 - Designed and created ML pipeline using AWS technologies such as SageMaker, S3, CloudWatch and IAM to enable MessageUs team to train, test and deploy machine learning models in a CI/CD fashion.
 - Redesigned the Device Item Picker in MessageUs, improving reliability and enabling customers to view and select their registered devices for support.
 - Participated in team's oncall rotations, authored correction of error (COE), and two-time hackathon winner.
- **Neural Systems Engineering and Information Processing Lab (NSEIP)** Los Angeles, USA
Research Assistant *Sept 2019 - Dec 2021*
 - Modelled large-scale brain networks during direct electrical stimulation, and developed nonlinear models that are amenable for control in closed-loop brain computer interface (BCI) applications.
- **Agency for Science, Technology and Research (A*STAR)** Singapore
Data Science Intern *Jun 2018 - Sept 2018*
 - Investigated epileptic seizure detection using a minimal number of EEG channels and developed a Convolutional Neural Network (CNN) architecture **SeizNet**.

TECHNICAL SKILLS

Programming Languages: Java, Scala, JavaScript, TypeScript, Python, MATLAB, C++, SQL

Machine/Deep Learning: Large language models (LLMs), Tensorflow, Scikit-learn, AutoGluon, NumPy, SciPy

AWS Technologies AWS CDK, CloudFormation, S3, Lambda, ECS, ECR, IAM, CloudWatch, Eventbridge, SQS, SNS, Clean Room, SageMaker, Bedrock

Others: Linux, Docker

EDUCATION

- **University of Southern California** Los Angeles, CA
Master of Science in Computer Science, CGPA: 3.96/4.00 *May 2022*
 - Selected Courses: Machine Learning, Analysis of Algorithms, Database Systems, Applied Machine Learning for Games, Estimation Theory, Random Processes, Applied Natural Language Processing
- **Bilkent University** Ankara, Turkey
Bachelor of Science in Electrical and Electronics Engineering, CGPA: 3.93/4.00 *Jun 2019*
 - Exchange student at Korea University for one semester

HONORS & AWARDS

- **Annenberg Fellowship with admission to University of Southern California (USC) PhD program, 2019**
- **Academic Excellence Graduation Award, 2019** Bilkent University Electrical and Electronics Engineering Department
- **IEEEExtreme Programming Competition**
 - **IEEEExtreme 10.0, 2016:** Ranked 2nd in Turkey, 20th in IEEE Region 8, and 65th overall
 - **IEEEExtreme 9.0, 2015:** Ranked 3rd in Turkey, 30th in IEEE Region 8, and 82nd overall
- **Nationwide University Entrance Exam (LYS):** Ranked in top 0.01% among 2 million students in Turkey, 2014

PUBLICATIONS

M. T. Avcu, Z. Zhang, and DWS. Chan "Seizure Detection Using Least EEG Channels by Deep Convolutional Neural Networks" IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019